

Day 4 - Largest of Two Numbers

Question

Do numbers diye gaye hain.

Dono me se bada number find karo.

Example

Input

```
a = 10  
b = 20
```

Output

```
20
```

Approach 1 - Using if-else

Logic

Agar $a > b$ hai to a largest hoga.

Warna b largest hoga.

Dry Run

$a = 15$

$b = 9$

$15 > 9$ ✓

Largest = 15

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Java Code

```
public class Day4 {  
    public static void main(String[] args) {  
  
        int a = 15;  
        int b = 9;  
  
        if(a > b)  
            System.out.println(a);  
        else  
            System.out.println(b);  
    }  
}
```

Approach 2 - Using Ternary Operator

Logic

Short form of if-else.

Java Code

```
public class Day4 {  
    public static void main(String[] args) {  
  
        int a = 10;  
        int b = 20;  
  
        int largest = (a > b) ? a : b;  
  
        System.out.println(largest);  
    }  
}
```

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Approach 3 - Using Math.max()

Logic

Java me built-in function available hai.

Java Code

```
public class Day4 {  
    public static void main(String[] args) {  
  
        int a = 25;  
        int b = 30;  
  
        System.out.println(Math.max(a, b));  
    }  
}
```

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Approach 4 - Without if-else (Interview Trick)

Logic

Formula:

$$\text{Largest} = (a + b + |a-b|) / 2$$

Example

a = 20

b = 15

Largest

$$= (20 + 15 + |20-15|) / 2$$

$$= (35 + 5) / 2$$

$$= 20$$

Java Code

```
public class Day4 {  
    public static void main(String[] args) {  
  
        int a = 20;  
        int b = 15;  
  
        int largest = (a + b + Math.abs(a - b)) / 2;  
  
        System.out.println(largest);  
    }  
}
```

Time Complexity

O(1)

Space Complexity

O(1)

Edge Cases

Case 1

a = 10
b = 10

Output

10

Dono same hain.

Case 2

a = -5
b = -10

Output

-5

Case 3

a = 0
b = 100

Output

100

Interview Follow-Up Questions

Q1. Math.max() internally kya karta hai?

Internally comparison hi karta hai.

```
return (a >= b) ? a : b;
```

Q2. Agar dono numbers same ho to?

```
a = 50  
b = 50
```

Largest = 50

Q3. Interview me best solution?

Pehle:

```
if (a > b)
```

batana.

Fir:

```
Math.max(a, b)
```

bhi mention karna.

Summary

Approach	Time	Space
if-else	O(1)	O(1)
Ternary	O(1)	O(1)
Math.max()	O(1)	O(1)
Formula Method	O(1)	O(1)

Interview Concept of the Day

Comparison Operators

>
<
>=
<=
==
!=

Ye operators DSA me bahut use honge:

- Searching
- Sorting
- Binary Search
- Trees
- Graphs

sabki foundation yahi hai.

Practice Questions

Predict Output:

1

```
int a = 50;  
int b = 100;
```

2

```
int a = -20;  
int b = -5;
```

3

```
int a = 0;  
int b = 0;
```

4

```
int a = 999;  
int b = 111;
```

5

```
int a = -100;  
int b = 100;
```


